3/8/26

PRACTICAL 4

Setting Up a local Area Network (LAN)

AIM:

To design, set up and configure a local Area Network (LAN) connecting four computers through a network switch, assign static IP addresses to all devices, verify connectivity using the ping command, and demonstrate resource/data sharing across the network.

IMPLEMENTATION IN CISCO PACKET TRACER:

Since physical hardware was limited, the LAN was also implemented and verified using CISCO packet tracer simulation server, following the steps below.

Step 1: Place the Devices

1. Open the Cisco Packet Tracer
2. From the device category panel, select End Devices and drag 4 PCs onto the workspace.
3. Select Switches and drag 1 switch (eg., 2960 series) onto the workspace.

Step 2: Cable the Devices

4. Click connections → select Copper - Straight - Through cable.
5. Click each PC, then click the switch, and select an available FastEthernet port to complete each connection.
6. Repeat for all four PCs (ports Fa0/1 to Fa0/4). Green link lights confirm a successful physical connection.

Step 3: Assign IP Addresses to PCs

7. Click a PC → Desktop tab → IP configuration.

8. Enter the static IP Address and Subnet Mask as per the IP table (Section 5) for each PC.

Step 4: Configure the Switch Management IP (optional)

9. Click the switch → CLI tab.

10. Enter the following commands to assign a management IP:

```
enable
configure terminal
interface vlan 1
ip address 10.1.1.5 255.0.0.0
no shutdown
exit
```

Step 5: Verify connectivity using Ping

11. Click a PC → Desktop tab → Command Prompt.

12. Use the ping command to test connectivity to the other PCs, eg., ping 10.1.1.2.

13. Successful "Reply from..." messages confirm connectivity.

Step 6: Demonstrate Data sharing (Server simulation)

14. Add a Server device (End devices → Server-PT) and connect it to the switch.

15. Assign it an IP address (eg., 10.1.1.10, subnet mask 255.0.0.0) via its desktop → IP configuration tab.

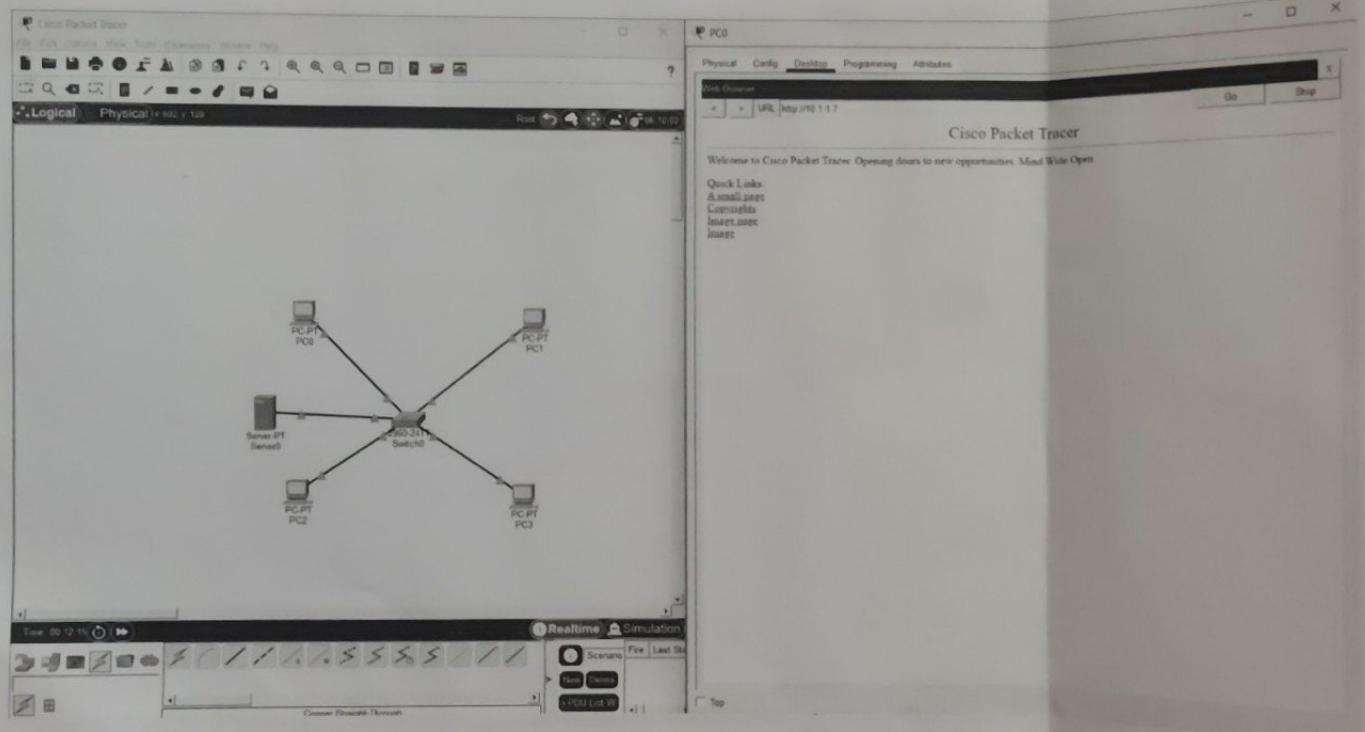
16. Open the server's services (config) tab and enable the HTTP service.

17. From another PC, open the Web Browser (Desktop tab) and enter the server's IP address to load the page, confirming successful data transfer across the LAN.

Step 7: Observe Packet Flow in Simulation Mode

18. Switch to Simulation mode (bottom-right of the workspace).

19. Send a ping or web request between the two devices and use the Play/Capture controls to step through the packet as it travels hop by hop through the switch, observing MAC-address-based forwarding.



Result:

All four PCs were successfully connected to the switch in a star topology.